

Smart Shock Detection & Automatic Power Cutoff System

Preventing Electrocution During Floods and Waterlogging

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Introduction

Floods and heavy rainfall are very common in India, especially during the monsoon season.

During floods, water often comes into contact with:

- Damaged electrical wires
- Broken electric poles
- Underground cables

This causes leakage current in waterlogged areas, creating a serious danger for:

- Pedestrians
- Residents
- Domestic animals

The proposed Smart Shock Detection system automatically disconnects power supply before accidents occur, providing a critical safety layer.

Key Features

Leakage Detection

Real-time sensing of current in waterlogged areas.

Automatic Cutoff

Instant disconnection of power supply upon threat.

Emergency Warning

Alerts via alarms and GSM communication.

Practical Implementation

Low-cost solution for urban safety.

Problem Statement

During floods and heavy rain, electrical safety becomes a major issue. Waterlogged areas may contain:

- Exposed live wires
- Damaged electrical systems
- Leakage current in standing water

Major Problems

- Exposed Wires
- Leakage Current
- Instant Public Risk
- Lack of Automatic Protection

Therefore, an intelligent and automated electrical safety system is required.

Main Objective

To develop a smart and automatic safety system that reduces electrocution risks during floods and waterlogging situations.

Specific Objectives

- Detect electrical leakage current
- Continuous monitoring of dangerous current levels
- Automatic power ON/OFF system
- Develop a low-cost portable and user-friendly device
- Alert public using alarms and indicators
- Enhance public safety during monsoon season

Proposed Solution

The proposed system is a **Smart Shock Detection and Automatic Power Cutoff System** designed to identify dangerous electrical leakage in flooded areas.

The system uses sensors to continuously monitor electrical leakage in waterlogged regions.

When leakage exceeds safe limits:

- Relay module disconnects power supply
- Buzzer produces warning alarm
- LEDs indicate danger status

System Advantages

- Fast Response Time
- Automatic Operation
- Improved Public Safety
- Flood-Prone Suitability

Components Required

Component	Function
Arduino UNO / ESP32	Main controller of the system
ACS712 Current Sensor	Detects leakage current
Water Level Sensor	Detects flood or water presence
Relay Module	Disconnects power supply automatically
Buzzer	Produces warning sound
LEDs	Shows system status
Battery Backup	Provides emergency power

- ① Sensor Detection
- ② Arduino Processing
- ③ Alert Generation
- ④ Safety Action

Workflow Process

- Sensors monitor environmental conditions
- Arduino processes sensor data
- Alert system generates warning
- Relay disconnects power supply

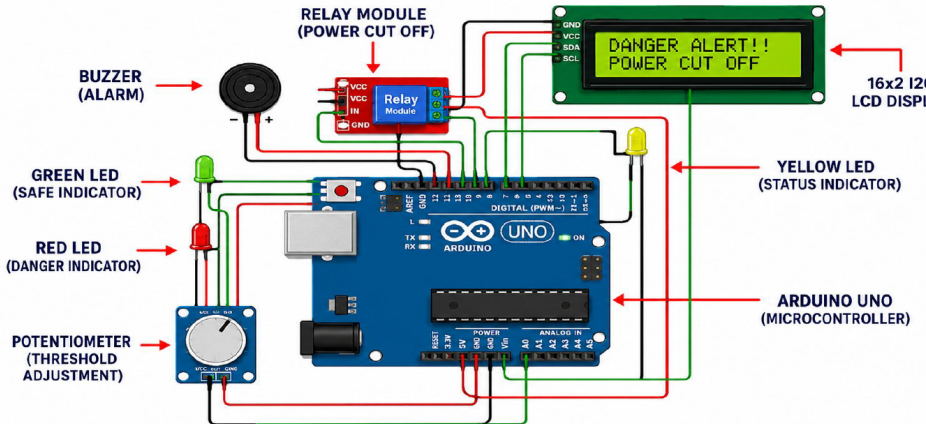
Working Principle

- 1 Detection using sensors
- 2 Processing by Arduino
- 3 Identification of threshold violations
- 4 Relay disconnects power supply
- 5 Buzzer and LEDs generate warning
- 6 Continuous monitoring cycle

The system ensures electrical isolation based on real-time leakage detection.

Circuit Diagram

SMART SHOCK DETECTION AND ELECTRICAL SAFETY ALERT SYSTEM



Contribution of Each Component

- **Arduino UNO** – Controls the entire system
- **Water Sensor** – Detects flood water
- **Current Sensor** – Monitors leakage current
- **Relay Module** – Disconnects power automatically
- **Buzzer** – Warns nearby people
- **LEDs** – Indicates safe/danger condition
- **Display** – Shows system status

Technologies Used

Category	Technologies
Hardware Technologies	Embedded Systems, Electrical Sensors, Relay Prot
Software Technologies	Arduino IDE, Embedded C Programming, Sensor I

SOFTWARE AND PROGRAMMING

SOFTWARE USED:



Arduino IDE



Embedded
C/C++

PROGRAM FUNCTIONS:

- ✓ Read sensor values
- ✓ Detect danger threshold
- ✓ Activate alarms
- ✓ Control relay cutoff
- ✓ Display warning message

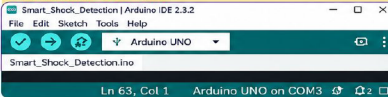
SMART FEATURES:

- Real-time monitoring
- Automatic response
- Fast processing

ARDUINO CODE

```
1 #include <Wire.h>
2 #include <LiquidCrystal_I2C.h>
3
4 LiquidCrystal_I2C lcd(0x27, 16, 2);
5
6 const int currentSensor = A0; // Current sensor (leakage detection)
7 const int waterSensor = A1; // Water sensor
8 const int buzzer = 8;
9 const int redLED = 7;
10 const int greenLED = 6;
11 const int relayPin = 5;
12 const int threshold = 500; // Danger threshold
13
14 void setup() {
15   pinMode(buzzer, OUTPUT);
16   pinMode(redLED, OUTPUT);
17   pinMode(greenLED, OUTPUT);
18   pinMode(relayPin, OUTPUT);
19
20   digitalWrite(relayPin, HIGH); // Relay ON (Power Connected)
21   digitalWrite(buzzer, LOW);
22   digitalWrite(redLED, LOW);
23   digitalWrite(greenLED, HIGH); // Safe indicator ON
24
25   lcd.init();
26   lcd.backlight();
27   lcd.setCursor(0, 0);
28   lcd.print("SYSTEM READY");
29   lcd.setCursor(0, 1);
30   lcd.print("SAFE MODE");
31 }
32
33 void loop() {
34   int currentValue = analogRead(currentSensor);
35   int waterValue = analogRead(waterSensor);
36
37   // Danger condition
38   if (currentValue > threshold || waterValue > 400)
39   {
40     digitalWrite(greenLED, LOW);
41     digitalWrite(redLED, HIGH);
42     digitalWrite(buzzer, HIGH);
43     digitalWrite(relayPin, LOW); // Relay OFF (Power Cutoff)
44
45     lcd.clear();
46     lcd.setCursor(0, 0);
47     lcd.print("DANGER ALERT!!!");
48     lcd.setCursor(0, 1);
49     lcd.print("POWER CUT OFF");
50   }
51   else {
52     digitalWrite(greenLED, HIGH);
53     digitalWrite(redLED, LOW);
54     digitalWrite(buzzer, LOW);
55     digitalWrite(relayPin, HIGH); // Relay ON
56
57     lcd.clear();
58     lcd.setCursor(0, 0);
59     lcd.print("SAFE MODE");
60     lcd.setCursor(0, 1);
61     lcd.print("MONITORING...");
62   }
63   delay(500); // Fast processing
64 }
```

ARDUINO IDE INTERFACE



The system continuously reads sensor data, processes it instantly and takes automatic action to ensure electrical safety.

Advantages

- Reduces accidental electrocution
- Automatic operation
- Fast response
- Low-cost implementation
- Easy installation
- Scalable for smart city applications
- Reliable protection system

Public Infrastructure

- Flood-prone roads
- Railway stations
- Bus terminals

Industrial Applications

- Electrical substations
- Construction sites
- Industrial safety zones

Future Enhancements

- IoT-based monitoring system
- Solar-powered emergency backup
- Cloud-based data storage
- AI-based danger prediction
- Smart electric grid connectivity
- Mobile application integration
- GPS-based danger tracking
- Automatic SMS alert system

Expected Outcome

- Reduction in electrocution accidents
- Faster emergency response
- Automatic power isolation
- Better monitoring of dangerous areas
- Improved public awareness

The project demonstrates how automation and embedded systems can solve practical real-life problems effectively.

Conclusion

The Smart Shock Detection and Automatic Power Cutoff System is an innovative and practical solution designed to reduce electrocution risks during floods.

Integrated Technologies

- Sensors & Automation
- Embedded Systems
- Electrical Protection Technologies

This system can play an important role in improving public safety and reducing electrical accidents in flood-prone areas.

Thank You

“Technology should not only make life easier, but also safer.”